

Michael Shi

Edmonton, AB | wenjian2@ualberta.ca | mshi.ca | linkedin.com/in/wjshi | github.com/micsushi

Education

University of Alberta, BSc in Computer Science with Specialization

Expected Graduation: **Dec 2026**

- **Awards:** Undergraduate Leadership Award, 4x Faculty of Science Scholarship, Dean's Honor Roll

Experience

Junior Software Developer (Part-time), INVIDI Technologies - Edmonton, AB

May 2025 - Feb 2026

- Enhanced real-time subscriber targeting accuracy by developing Kotlin microservices that integrated external marketing, forecasting, and analytics platforms via RESTful APIs.
- Established S3 cost saving blueprint for **100+** developers, resulting in a **74% (\$18,000) reduction** by implementing intelligent-tiering and S3 life cycle policies in our Terraform architecture.
- Accelerated deployment cycles and saved 2 hours of weekly developer time by automating CI/CD via Bitbucket pipelines and ECR/Kubernetes.
- Optimized bug detection speed by configuring Datadog metrics, monitors and centralized logging with automated alerting and error traces.
- Drove technical alignment by presenting completed features to stakeholders and leading design discussion meetings to gather feedback and refine system architecture.
- Minimized product downtime by resolving critical production node failures and bugs during on-call incident management.

Software Developer Intern, INVIDI Technologies - Edmonton, AB

May 2024 - May 2025

- Slashed backend operational costs by **89% (\$38,000)** and eliminated false-positive alerts by migrating the database from Apache Ignite to AWS Aurora PostgreSQL.
- Boosted authentication performance by **30%** by decoupling OAuth permissions into SPA and M2M clients across TypeScript/Vue.js and Kotlin services.
- Improved system performance and reduced API costs by implementing forecast caching in sync services using DynamoDB, eliminating redundant external calls.
- Maintained consistent team velocity during leadership transitions by assuming Scrum Master duties and facilitating Agile ceremonies and sprint planning.

Undergraduate Teaching Assistant, University of Alberta - Edmonton, AB

Sep 2023 - Present

- Taught multiple upper-level Computer Science courses, providing instructional support and mentoring students in advanced programming concepts.
- Mentored **500+** students in technical best practices, ensuring a **95% project success rate** in attaining A/A+ grades through Agile workflow facilitation.
- Reduced evaluation time by **85%** and ensured grading consistency by developing Python-based automation scripts for grading workflows.

Projects

StudyAmp: NatHacks 1st Place

github.com/NatRunners/StudyAmp

- Secured **\$4,000** in research funding by leading a team of 5 to develop an EEG-based attention monitoring web app during a hackathon.
- Optimized system scalability to support **10,000+** concurrent users by engineering a full-stack architecture on Vercel and Supabase.
- Achieved **85% accuracy** in attention scoring by developing a Python backend for real-time brainwave processing and data optimization.
- Enhanced user engagement by building a responsive UI using Next.js and Framer Motion based on beta tester feedback.
- Slashed recording review time by **80%** by integrating Gemini AI to automatically summarize audio content during detected attention drops.

Technical Skills

Languages: Python, C, C++, Java, Kotlin, R, JavaScript, TypeScript, Bash

Frameworks: Node.js, REST API, Express.js, Flask, FastAPI, React, Next.js, Vue.js, Tailwind

Developer Tools: Git, Docker, Kubernetes, MongoDB, Firebase, PostgreSQL, AWS, Gradle, Terraform, Datadog, Snyk